README.md — Complete Environment Setup Guide

Introduction:

This repository provides scripts and instructions to deploy the backend infrastructure needed for the **AI-Powered Product Finder** system across:

- **AlloyDB**
- **Cloud SQL PostgreSQL**
- **Cloud SQL MySQL**

This solution demonstrates intelligent search on a fashion product catalog using capabilities such as:

- Semantic vector queries
- Hybrid relevance search (Keyword + Vector search)
- Natural-language-to-SQL (AlloyDB exclusive)
- ScaNN vector indexing
- AI-powered query operators

The environment setup includes:

- GCS bucket creation
- SQL instance creation
- Schema and table creation
- VM creation (AlloyDB)
- Configuration setup
- Dataset loading & Pre-SQL transformations

Prerequisites:

Before starting, ensure the following.

Accounts & Access

- GitHub account
- GCP account with **Admin** permissions
- Cloud SQL MySQL admin access
- Cloud SQL PostgreSQL admin access
- AlloyDB PostgreSQL admin access
- A GCS bucket created in the format:

alloydb-usecase/search-usecase

Configurations

You must update required variables in the environment config files such as:

- PROJECT_ID
- REGION
- DB_PASSWORD
- BUCKET_NAME
- FOLDERS (e.g., raw/forecast, raw/ecomm, raw/eda)
- HOMEDIR
- CLONE_DIR
- CLONE_DIR_ECOMM
- CLUSTER_ID, INSTANCE_ID
- MACHINE_TYPE (e.g., n2-highmem-2)
- NETWORK_NAME
- ACCOUNT
- LOCATION

Cloud SQL MySQL Parameters

database-version = MYSQL_8_0_36
tier = db-n1-standard-1
edition = ENTERPRISE
enable-google-ml-integration = true

Cloud SQL PostgreSQL Required Roles

Assign the following roles to the service account:

- aiplatform.user
- vertexai.user

Installation:

Clone or Download Scripts

Place all setup scripts under the following path in Cloud Shell
<home_directory>/alloydb_gc/agentic/script

Source Dataset Reference

https://github.com/ldap/srcdump/tree/main/Ecommerce/dataset

Scripts Included

Common Scripts

- agentic_config.sh
- bucket_create.sh

AlloyDB Scripts

- alloydb_postgres_cluster_creation.sh

- ecomm_fashion_create_vm_inst.sh
- ecomm_fashion_wrapper_ddl.sh
- ecomm_fashion_ddl.sql
- ecomm_fashion_load_data_alloydb.sh
- ecomm_fashion_pre_sql.sh
- ecomm_fashion_presql_inst.sh
- ecomm_fashion_presql.sql

Cloud SQL MySQL Scripts

- cloudsql_mysql_instance_creation.sh
- ecomm_fashion_mysql_create_table.sql
- ecomm_fashion_mysql_create_ddl.sh
- ecomm_fashion_mysql_load_data.sh
- ecomm_fashion_mysql_presql.sql
- ecomm_fashion_mysql_presql_ddl.sh

Cloud SQL PostgreSQL Scripts

- cloudsql_postgres_instance_creation.sh
- ecomm_fashion_cloudsql_create_table.sql
- ecomm_fashion_cloudsql_create_ddl.sh
- ecomm_fashion_cloudsql_load_data.sh
- ecomm_fashion_cloudsql_presql.sql
- ecomm_fashion_cloudsql_presql.sh

Usage:

Below is the unified runbook for all three databases.

Step 0 – Create Required GCS Bucket Structure

Run: bucket_create.sh
This creates the bucket + folders required to store datasets.

Step 1 – Create Database Instances

MySQL

Run: cloudsql_mysql_instance_creation.sh

PostgreSQL

Run: cloudsql_postgres_instance_creation.sh

AlloyDB

Run: alloydb_postgres_cluster_creation.sh

Step 2 – Update Database Password (For Cloud SQL MySQL)

Go to: GCP Console → Cloud SQL → Instances → Users

Change **root** password → update in: agentic_config.sh

Step 3 – Create Tables & Schema

MySQL

Run: ecomm_fashion_mysql_create_ddl.sh
Executes: ecomm_fashion_mysql_create_table.sql

PostgreSQL

Run: ecomm_fashion_cloudsql_create_ddl.sh
Executes: ecomm_fashion_cloudsql_create_table.sql

AlloyDB

Run: ecomm_fashion_create_vm_inst.sh

This:

- Creates a VM
- Executes ecomm_fashion_wrapper_ddl.sh → runs ecomm_fashion_ddl.sql
- Prompts encryption key → enter Alloydb
- Creates schema: alloydb_usecase

Step 4 – Upload Dataset to GCS

Place the dataset (fashion_dataset.csv) into your Git folder.

Run: ecomm_fashion_git_to_gcs.sh
This loads dataset into: alloydb-gc-usecase-newsetup/raw/ecomm

Step 5 – Load Data into Database

MySQL

ecomm_fashion_mysql_load_data.sh

PostgreSQL

ecomm_fashion_cloudsql_load_data.sh

AlloyDB

ecomm_fashion_load_data_alloydb.sh

Step 6 – Pre-SQL / Embedding Column Creation

MySQL

ecomm_fashion_mysql_presql_ddl.sh

Executes: ecomm_fashion_mysql_presql.sql

PostgreSQL

ecomm_fashion_cloudsql_presql.sh

AlloyDB

ecomm_fashion_presql_inst.sh

Generic Steps to Run Any Script in Google Cloud Shell:

1. Open Google Cloud Shell and run the following command: **gcloud auth login**
2. Press Enter. When prompted to continue, type Y and press Enter again.
3. A URL will be displayed. Click on the link, which will redirect you to the Google sign-in page.
4. Select the appropriate Google account, click Continue, and grant the required permissions.
5. After successful authentication, an authorization code will be generated.
6. Copy the code and paste it back into the Cloud Shell, then press Enter.
7. Finally, set the required account and project by running the following commands:
    - gcloud config set account "<Account Id>"
    - gcloud config set project "Project Id"

Folder Structure:

```
alloydb_gc/
 └── agentic/
     └── script/
         ├── agentic_config.sh
         ├── bucket_create.sh
         ├── cloudsql_mysql_instance_creation.sh
         ├── cloudsql_postgres_instance_creation.sh
         ├── alloydb_postgres_cluster_creation.sh
         ├── ecomm_fashion_mysql_create_table.sql
         ├── ecomm_fashion_mysql_create_ddl.sh
         ├── ecomm_fashion_cloudsql_create_table.sql
         ├── ecomm_fashion_cloudsql_create_ddl.sh
         ├── ecomm_fashion_wrapper_ddl.sh
         ├── ecomm_fashion_ddl.sql
         └── (all other scripts)
```